

Question Paper

Exam Date & Time: 31-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025
SOFTWARE PROJECT MANAGEMENT [IT_3125]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Consider an organization executing two projects simultaneously. One project focuses on developing a cloud-based e-learning platform, while the other involves establishing a physical data centre infrastructure to support enterprise clients. The projects have comparable budgets and timelines. (5)
- A) Apply Project Management principles to identify 3 major differentiations between these two projects in terms of scope, risk management, and resource allocation. Justify your answer.
- Course Outcome: "CO1" / Cognitive Level: "Application"
- B) A tech startup plans to design a smart home automation system requiring an initial investment of ₹20 lakh. The anticipated cash inflows for the next three years are ₹6 lakh, ₹8 lakh, and ₹10 lakh respectively. The management needs to assess whether to move ahead with the project. (3)
- Apply appropriate cost-benefit evaluation techniques to determine if the project is financially feasible.
- Specify the metric or approach used (e.g., NPV, IRR, or Payback Period) and justify your choice.
- Course Outcome: "CO1" / Cognitive Level: "Application"
- C) An enterprise embarks on a Sustainability and Digital Innovation Programme, consisting of projects in green data centre implementation, AI-driven energy optimization, and paperless workflow automation. The leadership team wants to ensure that each project is strategically aligned with the organization's long-term goals of environmental responsibility, cost efficiency, and digital agility. (2)
- Apply the concept of Strategic Assessment to explain how the management can evaluate the alignment of each project with the organization's strategic objectives, clearly illustrating the use of SMART objectives in this assessment.
- Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) A hospital's IT department plans to develop an automated patient record management system to integrate data from multiple departments such as laboratories, pharmacy, and billing. However, unclear requirements and uncoordinated communication among stakeholders are causing confusion and delays in project initiation. Apply the Stepwise Project Planning framework to demonstrate how the project manager can effectively establish the steps in the project planning process. (5)
- A)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- B) Assume that the size of an organic type of software product has been estimated to be 32,000 lines of source code. Assume that the average salary of software engineers is Rs. 15,000/- per month. Determine the effort required to develop the software product and the nominal development time. [a

= 2.4, b = 1.05, c = 2.5, d = 0.38]

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) A startup is developing a mobile health tracking app for users whose requirements keep evolving during testing. The team wants faster feedback and early delivery of usable features. Apply the concept of Agile methodology to explain how this approach helps the team manage changing requirements effectively while ensuring customer satisfaction. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) A software firm is developing an online learning management system involving multiple interdependent activities such as requirement analysis, UI design, coding, testing, and deployment. (5)
- A) During project execution, the manager identifies delays in the design and testing phases. Apply the Critical Path Method (CPM) to explain how the project manager can determine which activities are critical, calculate the slack time, and revise the schedule to ensure timely project completion. Discuss how CPM aids in prioritizing resource allocation and controlling project delays.

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) A university software project team is developing a centralized student management portal. The same database specialist is assigned to both backend integration and reporting modules. Midway through the project, both modules require his attention simultaneously, causing schedule delays. Analyse the resource allocation issue in this scenario and suggest how the project manager can identify and resolve such resource bottlenecks using effective scheduling or resource levelling techniques. (3)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- C) During the development of an AI-based diagnostic system, the project team faces uncertainty regarding the accuracy of a new machine-learning model and potential delays in data acquisition. Apply the risk management process to suggest how the team can identify, assess, and plan mitigation strategies for these uncertainties. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 4) A multinational software firm is implementing an Enterprise Resource Planning (ERP) system across its global branches. As the project progresses, the manager notices inconsistent status updates, overlapping responsibilities, and lack of visibility into inter-team dependencies, leading to coordination issues and schedule slippages. (5)
- A) Apply an appropriate Project Management and Control framework to illustrate how the project manager can:

- Establish structured monitoring and reporting mechanisms,
- Define clear accountability through responsibility assignment matrices (like RACI), and
- Implement periodic review and change control processes to ensure alignment with project goals.

Explain how applying this framework strengthens project governance, enhances team coordination, and supports timely project completion.

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) A software firm is developing a cloud-based HR management system. During development, frequent feature enhancement requests from stakeholders and simultaneous updates by multiple developers have resulted in inconsistent builds and integration issues. (3)

Apply the principles of Change and Software Configuration Management (SCM) to explain how the project manager can:

- Establish a formal change control process for evaluating and approving modifications,

- Implement version control mechanisms to manage concurrent development, and

Ensure traceability and consistency across all project components.

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) An educational institution partners with a software vendor to develop a learning analytics dashboard (2) under a time-and-material (T&M) contract. Explain how a time-and-material contract provides flexibility in managing project scope changes while maintaining transparency between the client and vendor.

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 5) A corporate company has a team of 8 employees working on a critical client project. Recently, (5)
conflicts have arisen due to overlapping responsibilities and communication gaps. Deadlines are being missed, and client satisfaction is at risk.
- A)

Analyse the team's current challenges and identify the key reasons for conflict and inefficiency. Suggest strategies to improve team collaboration and performance, supporting your recommendations with logical reasoning.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Evaluate the efficacy of the Oldham-Hackman Job characteristic Model in motivating the developer (3) community while performing mundane tasks.

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- C) Explain why formal acceptance from the client is required before closing a project. (2)

Course Outcome: "CO4" / Cognitive Level: "Understand"

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Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	IT_3125:SOFTWARE PROJECT MANAGEMENT

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	20
3	3	CO 3	10
4	4	CO 4	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	IT_3125:SOFTWARE PROJECT MANAGEMENT

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	4(8%)
3	Application	9	33(66%)
4	Analysis	3	10(20%)
5	Create	1	3(6%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)

